

4-2-1: Examining IPv4 Address Classes

At the end of this episode, I will be able to:

1. Identify and explain the structure of IPv4 address classes and importance, given a scenario.

Learner Objective: *Identify and explain the structure of IPv4 address classes and importance, given a scenario*

Description: In this episode, the learner will examine the IPv4 addressing class structure. We will explore classes ranges, network and hosts capacities, common usage and more.

- What are the address ranges of each IPv4 class?
 - o Class A - 1.0.0.0 to 126.0.0.0 (1.0.0.0/8 to 126.0.0.0/8)
 - o Class B - 128.0.0.0 to 191.255.0.0 (128.0.0.0/16 to 191.255.0.0/16)
 - o Class C - 192.0.0.0 to 223.255.255.0 (192.0.0.0/24 to 223.255.255.0/24)
 - o Class D - 224.0.0.0 to 239.255.255.255
 - o Class E - 240.0.0.0 to 255.255.255.255
 - How many networks and hosts can each IPv4 class support?
 - o Class A - Supports 128 networks with approximately 16 million hosts each
 - o Class B - Supports 16,384 networks with 65,534 hosts each
 - o Class C - Supports over 2 million networks with up to 254 hosts each
 - o Class D - Not used for standard host addresses (multicast)
 - o Class E - Reserved for experimental; not for general use
 - What are the common uses for each IPv4 class?
 - o Class A - Large organizations or networks requiring a vast number of IP addresses
 - o Class B - Medium-sized networks such as universities or regional services
 - o Class C - Small businesses and local area networks (LANs)
 - o Class D - Multicast groups
 - o Class E - Experimental purposes
 - Can you provide examples of where each IPv4 class is typically used?
 - o Class A - Major global enterprises, internet service providers
 - o Class B - Educational institutions, larger corporations
 - o Class C - Small companies, office networks
 - o Class D - Streaming services, multi-user games
 - o Class E - Network protocol testing, future use experimentation
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- Additional Resources:
 - o If applicable